Differential Geometry Assignment 1

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Let $\Psi = \Psi_p \in L(C^\infty(M), [T_p(M)]^*)$ be the derivative at $p \in M$
We have a local parametrization $\phi : B \to V$ with inverse φ

1 Bump functions

- We have $f \in C^\infty(\mathbb{R})$ given by $f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$
- Define $g \in C^\infty(\mathbb{R})$ by $g(x) = \frac{f(x)}{f(x)+f(1-x)}$. We have $g(x) = 0$ for $x \leq 0$ and $g(x) = 1$ for $x \geq 1$
- Define $h \in C^\infty(\mathbb{R}^n)$ by $h(x) = g(4-2||x||)$. We have $h(x) = 1$ for $||x|| \leq 1$ and $h(x) = 0$ for $||x|| \geq 2$

Claim 1. *Suppose $K \subseteq U \subseteq M$ where K is compact and U is open in M. Then there is $f \in C^\infty(M, [0,1])$ with support inside U such that $f(x) = 1$ for all $x \in K$*

Proof. There are multiple steps

- By local compactness of M, get $V \subseteq \overline{V} \subseteq U$.
- For each $x \in K$, get a local parametrization $\phi_x : B_3 \to V_x \subseteq V$. Define $f_x \in C^\infty(M, [0,1])$ by $f_x(q) = \begin{cases} h((\phi_x)^{-1}(q)) & \text{if } q \in V_x \\ 0 & \text{if } q \notin V_x \end{cases}$
- By compactness, get $x_1, x_2, \ldots, x_n$ such that V_{x_i} cover K. Let $f = 1 - \prod(1 - f_{x_i})$

$\square$

Claim 2. *Suppose $g \in C^\infty(U)$. Then there is $p \in V \subseteq U$ and $f \in C^\infty(M)$ such that $f_{|V} = g_{|V}$*

2 Point derivations

- Let $\text{Der}_p \subseteq [C^\infty(M)]^*$ denote the space of point derivations at p.
- Let $W = \bigcap_{B \in \text{Der}_p} \ker B$

Claim 3. *Suppose $f \in C^\infty(M)$ vanishes in some neighborhood of p. Then $f \in W$*

Claim 4. *Let $v \in T_pM$. Then $D_v \in Der_p$*

Claim 5. *We have $\tilde{\Psi}(J(v)) = D_v$, where $\tilde{\Psi}$ is the dual map and J is the canonical evaluation map.*

Claim 6. *We have $W = ker\ \Psi$*

Proof. We prove both inclusions separately.

- Suppose $df_p = 0$. By Hadamard's lemma there are $g_i \in C^\infty(B)$ such that

$$f(\phi(x))) = f(p) + \sum_{i=1}^{n} x_i g_i(x)$$

  Differentiating, we see that $g_i(0) = 0$. By Claim 2, it suffices to prove that $\sum_{i=1}^{n}(\varphi_i)[g_i \circ \varphi] \in W$. But this follows from the product property.
- The other direction follows from Claim 4

□

Claim 7. *Ψ is surjective*

Proof. Here are the steps

- Let $v_i = (D\phi)_0 e_i$ and g_i be the dual basis
- Get $f_i \in C^\infty(M)$ and $U \subseteq V$ such that $f_i(q) = \varphi_i(q)$ for all $q \in U$.
- Then $\Psi(f_i) = g_i$

□

Claim 8. *$\tilde{\Psi}$ is injective*

Claim 9. *$v \to D_v$ is injective*

Claim 10. *We have range$(\tilde{\Psi}) = Der_p$*

Proof. By Claim 2 we get an injective linear map from Der_p to $\left(\frac{C^\infty(M)}{W}\right)^*$, which by Claims 5 and 6 is isomorphic to T_pM. □

3 Derivations

Let Der denote the space of derivations.

Claim 11. *For any $B \in Der, B_p \in Der_p$*

Claim 12. *For any $X \in \mathbb{X}(M)$, we have $D_X \in Der$*

Define $\Phi : \mathrm{Der} \to \mathrm{Map}(M, \mathbb{R}^k)$ by $D_{\Phi(B)(p)} = B_p$

Claim 13. *We have range $\Phi \subseteq \mathbb{X}(M)$*

Proof. Get f_i as before. We have $B_p = \sum_{i=1}^n B_p f_i D_{\phi_i(p)}$ □

Claim 14. *The map $X \to D_X$ is a linear isomorphism onto Der with inverse Φ*

4 Regular curves

Claim 15. *There is a local parametrization $\phi : B \to V$ such that $(D\phi)_0$ is an isometry*

Proof. Let $\tilde{\phi} : B \to V$ be a local parametrization and let $v_1, v_2, \dots, v_n$ be an orthonormal basis of T_pM. We get $A \in GL(\mathbb{R}^n)$ such that $(D\tilde{\phi})_0 Ae_i = v_i$. Define $\phi : A^{-1}B \to V$ by $\phi = \tilde{\phi} \circ A$. □

Claim 16. *We can make $L(\eta) - d_M(p, q)$ arbitrarily small for a piecewise regular curve η from p to q*

Proof. Here are the steps

- Get a piecewise smooth curve γ from p to q
- By compactness, we may reduce to the case that there is a local parametrization $\phi : B \to V$ such that $\gamma \subseteq V$ and by Claim 8 we may assume $||A_x||, ||B_x|| \leq 1 + \delta$, where $A_x = (D\phi)_x$ and $B_x A_x = 1$
- Define $\eta(t) = \phi(t\varphi(q) + (1 - t)\varphi(p))$
- We have $L(\eta) \leq (1 + \delta)^2 L(\gamma)$

□